

Chapter 3 - System Analysis & Feasibility Study

3.1 Existing System

Many organizations manage customer support through spreadsheets, emails, or disconnected applications. Manual handling of customer information and service requests often results in duplicated records, delayed responses, inconsistent documentation, and higher operational effort. Traditional systems also lack intelligent automation for case routing and summary generation, making support processes slower and more error-prone.

3.2 Limitations of the Existing System

- Manual data entry increases the possibility of human errors.
- Customer cases are assigned manually, causing delays.
- Support summaries require manual preparation.
- Limited visibility into case progress.
- Inconsistent data validation.
- No AI-assisted support or intelligent automation.

3.3 Proposed System

The proposed solution is an AI-Powered Customer Support & Service Automation application developed on Salesforce. It centralizes customer, product, order, and support case information using custom objects. Validation Rules ensure accurate data entry, Record-Triggered Flows automate business processes, Queues route cases automatically, and Prompt Builder generates AI-powered summaries for customer support personnel.

The screenshot displays the Salesforce interface for a Customer record. At the top, there's a navigation bar with 'Developer Edition' and a search bar. Below this, a tabbed interface shows 'Customers' as the active tab, with other tabs like 'Merchandises', 'Orders', 'Use Cases', 'Reports', and 'Dashboards'. The main content area is titled 'Customer' and shows the name 'Vellampalli Venkata Nikhil AP24110010587'. To the right of the name are buttons for 'New Contact', 'Edit', and 'New Opportunity'. Below the name, there's a 'Details' tab with a table of customer information. The table has two columns: 'Customer Name' and 'Owner'. The 'Customer Name' column contains 'Vellampalli Venkata Nikhil AP24110010587' and 'Email' 'venkatanikhil.ap24110010587@srmmap.edu.in'. The 'Owner' column contains 'Vellampalli Venkata Nikhil AP24110010587'. Other fields include 'Phone', 'Customer Type' (Individual, Business), 'Created By' (Vellampalli Venkata Nikhil AP24110010587, 7/6/2026, 8:42 AM), and 'Last Modified By' (Vellampalli Venkata Nikhil AP24110010587, 7/6/2026, 8:42 AM). On the right side, there's an 'Activity' section with a 'Filters' dropdown set to 'All time • All activities • All types'. Below the filters, there's a 'Refresh' button and a 'Show All Activities' button. The activity list is currently empty, showing 'No activities to show. Get started by sending an email, scheduling a task, and more.'

Customer Name	Owner
Vellampalli Venkata Nikhil AP24110010587	Vellampalli Venkata Nikhil AP24110010587

Related: Details

Customer Name: Vellampalli Venkata Nikhil AP24110010587

Email: venkatanikhil.ap24110010587@srmmap.edu.in

Phone:

Customer Type: Individual, Business

Created By: Vellampalli Venkata Nikhil AP24110010587, 7/6/2026, 8:42 AM

Last Modified By: Vellampalli Venkata Nikhil AP24110010587, 7/6/2026, 8:42 AM

Activity

Filters: All time • All activities • All types

Refresh • Expand All • View All

Upcoming & Overdue

No activities to show. Get started by sending an email, scheduling a task, and more.

To change what's shown, try changing your filters.

Show All Activities

3.4 Advantages of the Proposed System

Feature	Benefit
Automation	Reduces manual effort and improves response time.
Validation Rules	Improves data accuracy.
Queues	Automatically routes customer cases.
Prompt Builder	Generates AI-assisted summaries.
Lightning UI	Provides a modern and user-friendly interface.
Scalability	Supports future enhancements and integrations.

3.5 Feasibility Study

3.5.1 Technical Feasibility

Salesforce Developer Edition provides all required technologies including Flow Builder, Prompt Builder, Lightning App Builder, Validation Rules, Queues, and Custom Objects, making the project technically feasible.

3.5.2 Economic Feasibility

The project is economically feasible because it uses the free Salesforce Developer Edition and requires only a computer with internet access.

3.5.3 Operational Feasibility

The Lightning interface is intuitive and easy to use. Automation reduces repetitive work and improves operational efficiency.

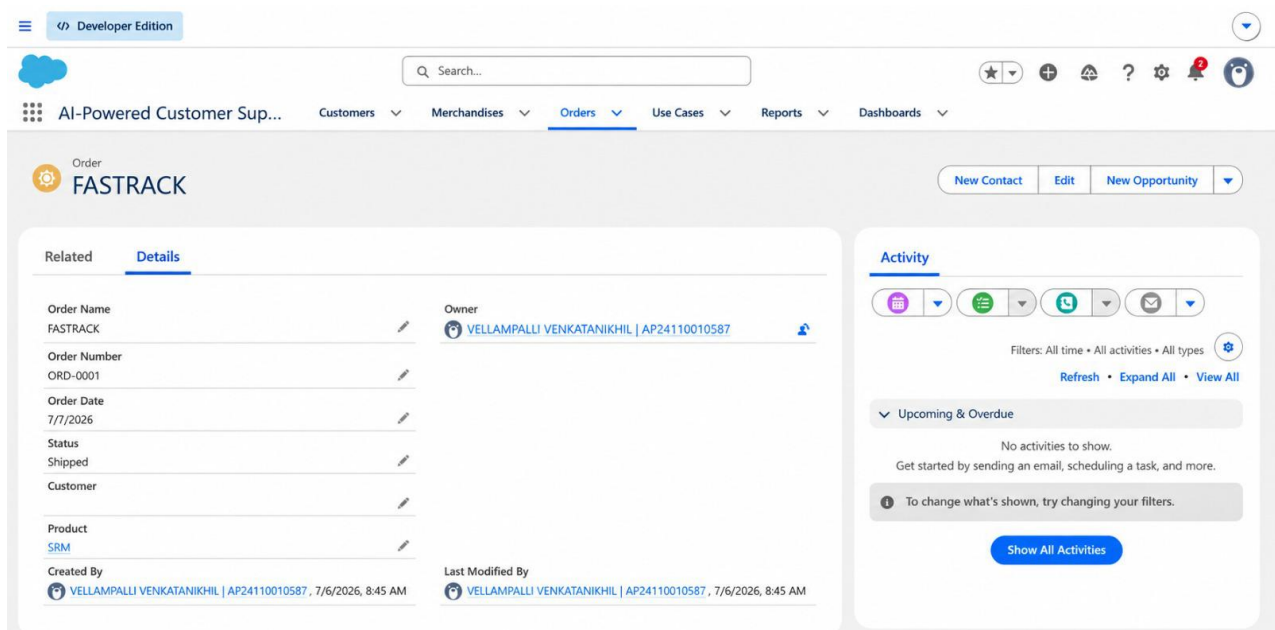
3.5.4 Schedule Feasibility

The project was completed within the planned schedule through phased implementation, testing, and deployment.

3.6 System Architecture

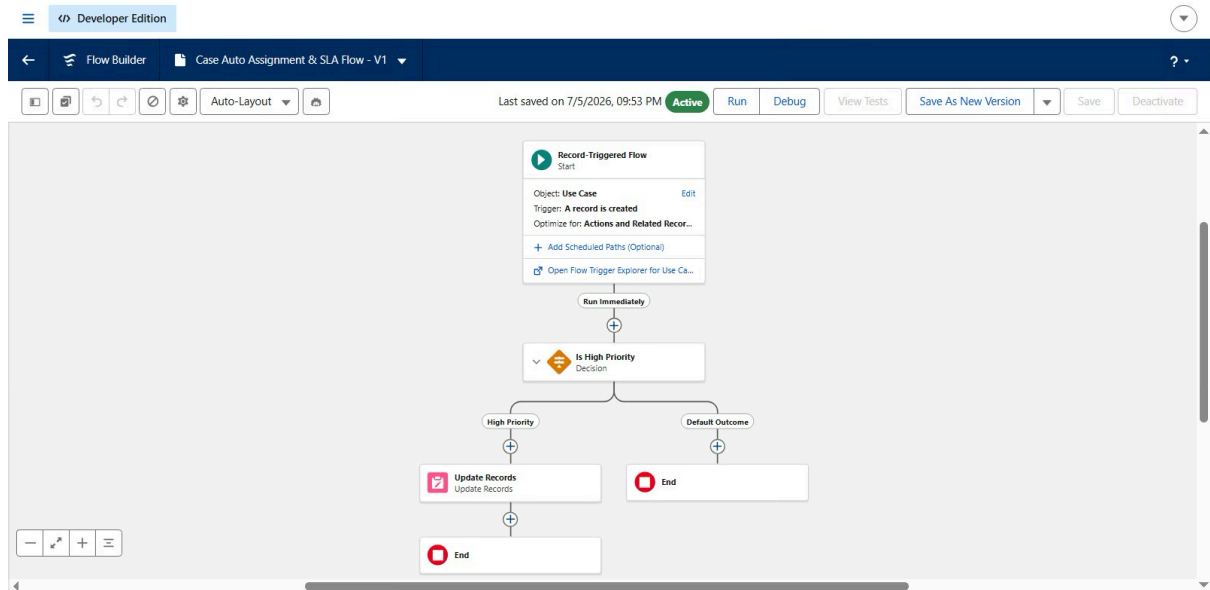
The solution follows a layered architecture:

- **Presentation Layer:** Salesforce Lightning UI
- **Application Layer:** Flows, Validation Rules, Prompt Builder
- **Data Layer:** Customer, Product, Order, and Use Case objects
- **Security Layer:** Profiles, Queues, and Field-Level Security
- **AI Layer:** Prompt Builder for AI-generated summaries



3.7 System Workflow

User Login → Lightning App → Customer / Product / Order → Use Case → Validation Rules → Record Triggered Flow → Queue Assignment → AI Summary Generation → Reports



3.8 Chapter Summary

The analysis confirms that the proposed Salesforce-based solution effectively overcomes the limitations of manual customer support systems. The feasibility study demonstrates that the project is technically, economically, operationally, and schedule-wise practical while providing an intelligent, scalable, and secure customer support platform.